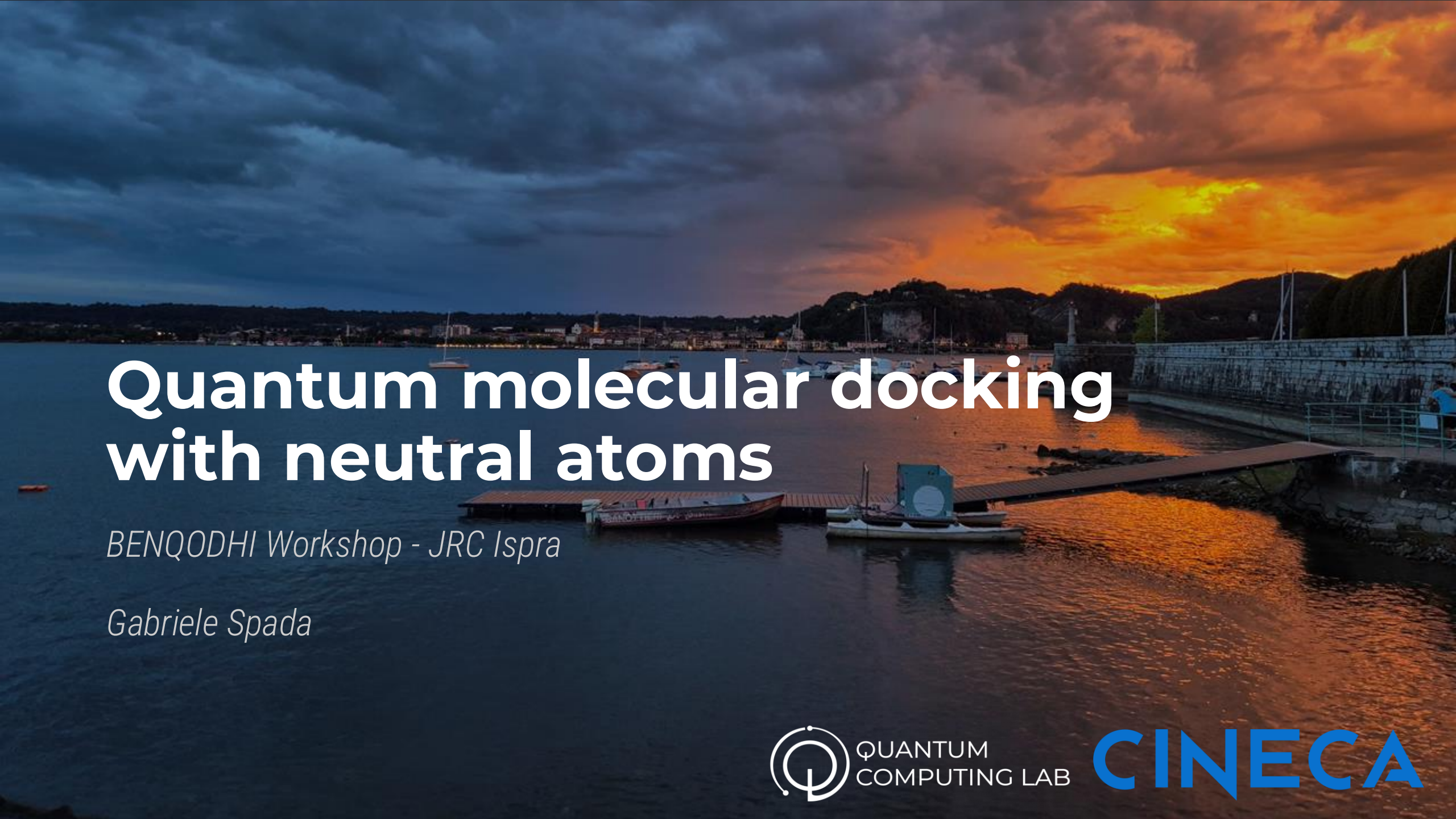




Quantum molecular docking with neutral atoms

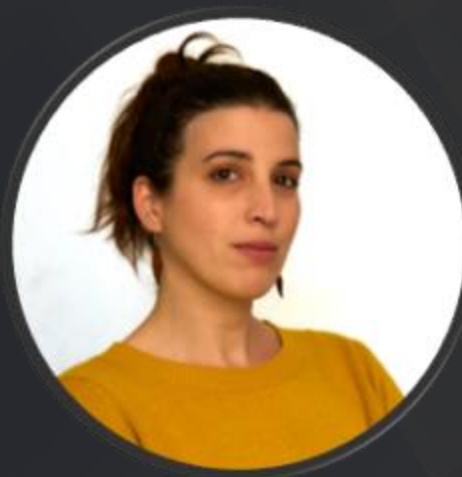
BENQODHI Workshop - JRC Ispra

Gabriele Spada

Cineca Quantum Computing Lab



Sara Marzella



Anita Camillini



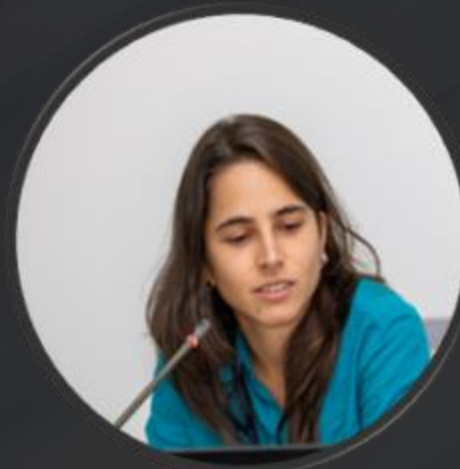
Vito Palmisano



Jonathan
Frassinetti



Gabriele Spada



Francesca Gebbia

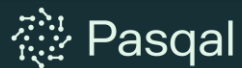


Antonio
Costantini

Quantum Computing Lab



- PASQAL Orion Beta 140 qubits
Analog Quantum Computer
- IQM Radiance Machine 54 qubits
Digital Quantum Computer
- Cloud Resources via ISCRA-C
D-Wave Quantum Annealers
Pasqal 100 qubits Quantum Simulator



European and National projects



PON QSUD



HPC4UPC



MoSeGad



CINECA



Quantum «EDU» - Nox



- Superconducting Digital Quantum Computer
- 54 qubits
- 90 couplers
- Installed, currently integrating with HPC

Cineca to house Italy's most powerful quantum computer IQM Radiance 54

17/03/2025

🕒 3 min. read

- IQM Radiance 54-qubit full-stack superconducting quantum computer will be integrated into Leonardo, one of the world's fastest supercomputers in Bologna, Italy.
- Cineca intends to use the system for optimisation of quantum applications, quantum cryptography, quantum communication and artificial intelligence quantum algorithms.

IQM **CINECA**

EuroQCS-Italy - Sol

- Installation in progress
- 140 qubits **analog** quantum computer
- Upgrade with **local addressability** next year



Quantum Excellence Centre for Quantum-Enhanced Applications

Horizon EUROHPC project designed as a pan-European one-stop shop for quantum-enhanced applications.

It aims to bridge the gap between QC developers and end users by offering:

Hardware-independent software libraries
Assisted access to European QC and HPC infrastructures
Development tools, training, and support

Organized around five geographic clusters of competence centres across Europe, each built around key HPC/QC hubs.

Italian coordination



CINECA	Lead Italian HPC center; EuroHPC JU QC host (EuroQCS-Italy); central hub of Cluster B.
E4 Computer Engineering	SME specialized in QC/HPC hybridization; co-developer of applications (e.g. drug discovery).
LINKS Foundation	Applies QC to optimization problems like MIS and scheduling.
University of Padova (UniPD)	Academic partner developing quantum algorithms and applications (PDEs, PQC, scheduling).



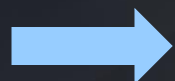
EuroHPC
Joint Undertaking

CINECA

 **QUANTUM
COMPUTING LAB**

CINECA's contributions:

- Operate quantum computing infrastructure (EuroQCS-Italy).
- Contribute to building a library of hardware-independent quantum applications.
- Support training and education, especially in application-oriented QC.
- Help design compilers/APIs for quantum-classical integration.
- Play a key role in sustainability and dissemination of QC expertise and tools.



First Application involving CINECA and Partners (E4, PSNC) Molecular Docking & Unfolding

Field: Quantum chemistry, drug discovery

Goal: Use quantum optimization (QAOA, QAA) for protein-ligand binding and unfolding.

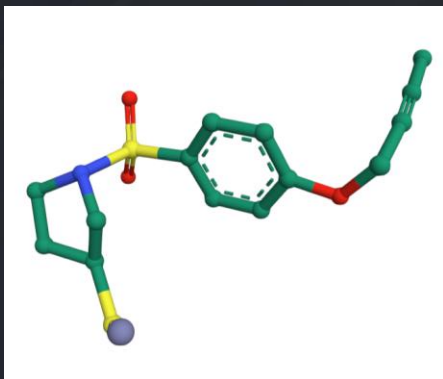
Outcome: Hybrid HPC-QC code building on experience from LIGATE and EuroCC projects.

Molecular Docking

Unbound 3D structure

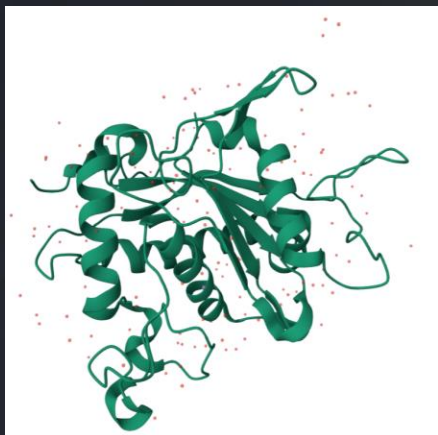
- Drug molecule (ligand)

Aryl-sulfonamide



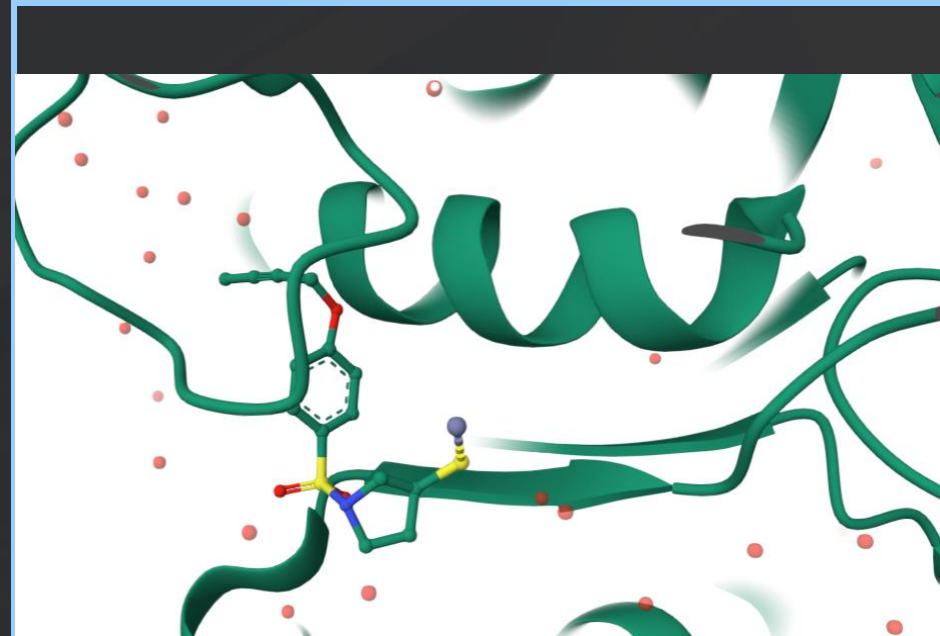
- Target macromolecule (receptor)

TNF- α Converting Enzyme



MD

3D orientations



+ scores

Quantum approach to MD

SCIENCE ADVANCES | RESEARCH ARTICLE

PHYSICAL SCIENCES 5 Jun 2020, Vol 6, Issue 23, [DOI: 10.1126/sciadv.aax1950](https://doi.org/10.1126/sciadv.aax1950)

Molecular docking with Gaussian Boson Sampling

Leonardo Banchi^{1*}, Mark Fingerhuth², Tomas Babej², Christopher Ing², Juan Miguel Arrazola¹

Gaussian Boson Samplers are photonic quantum devices with the potential to perform intractable tasks for classical systems. As with other near-term quantum technologies, an outstanding challenge is to identify specific problems of practical interest where these devices can prove useful. Here, we show that Gaussian Boson Samplers

Molecular docking via weighted subgraph isomorphism on quantum annealers

Emanuele Triuzzi, Riccardo Mengoni, Francesco Micucci, Domenico Bonanni, Daniele Ottaviani, Andrea Rosario Beccari and Gianluca Palermo*

Published 30 September 2025 • © 2025 The Author(s). Published by IOP Publishing Ltd

[Quantum Science and Technology, Volume 10, Number 4](#)

Citation Emanuele Triuzzi et al 2025 *Quantum Sci. Technol.* **10** 045049

DOI 10.1088/2058-9565/ae0890

Quantum molecular unfolding

Kevin Mato, Riccardo Mengoni, Daniele Ottaviani and Gianluca Palermo

Published 9 June 2022 • © 2022 The Author(s). Published by IOP Publishing Ltd

[Quantum Science and Technology, Volume 7, Number 3](#)

Citation Kevin Mato et al 2022 *Quantum Sci. Technol.* **7** 035020

DOI 10.1088/2058-9565/ac73af

arXiv:2604.09540v1 [cs.ET] 10 Apr 2026

A Physically-Informed Subgraph Isomorphism Approach to Molecular Docking Using Quantum Annealers

Francesco Micucci¹, Matteo Barbieri³, Gabriella Bettonte³, Domenico Bonanni⁵⁶

Anita Camillini⁴, Anna Fava², Daniele Gregori³, Andrea R. Beccari², Gianluca Palermo^{1,7}

arXiv:2508.18147v2 [quant-ph] 24 Apr 2026

A Scalable Heuristic for Molecular Docking on Neutral-Atom Quantum Processors

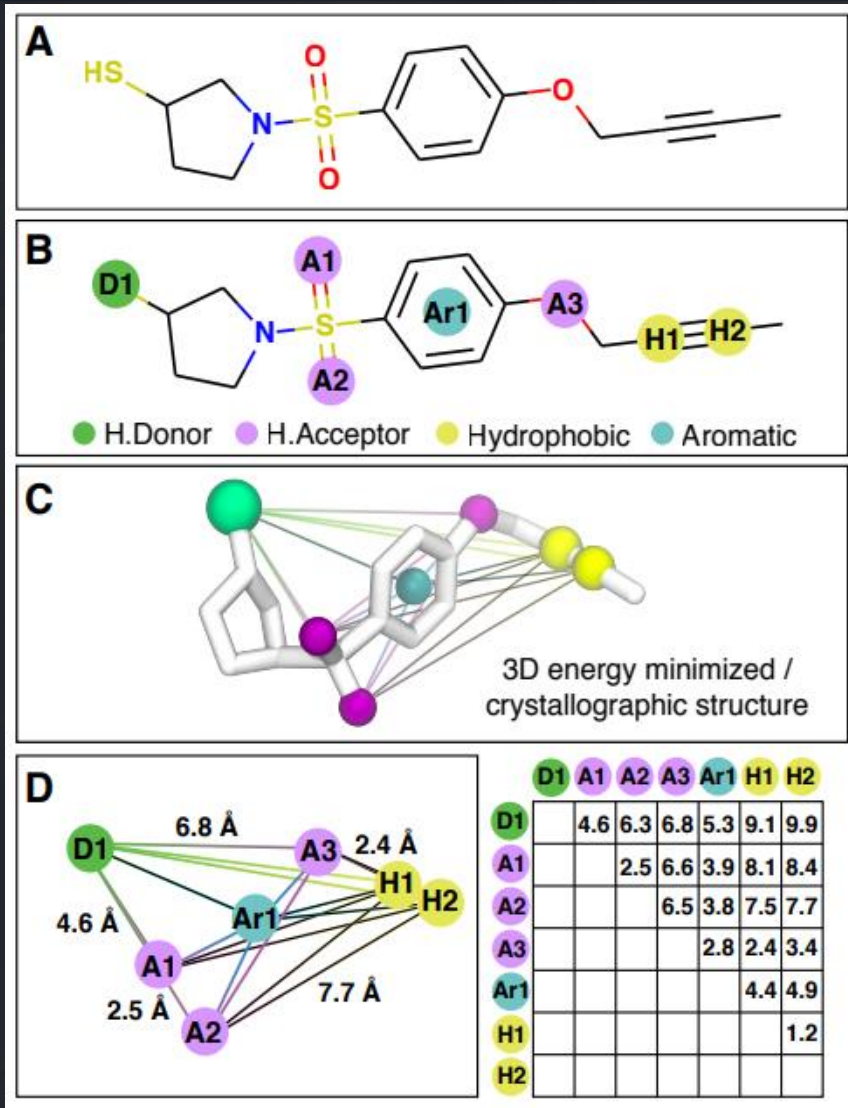
Mathieu Garrigues[†]

Victor Onofre[†]

Wesley Coelho

S. Acheche

Labelled distance graph



→ Planar structure of the ligand

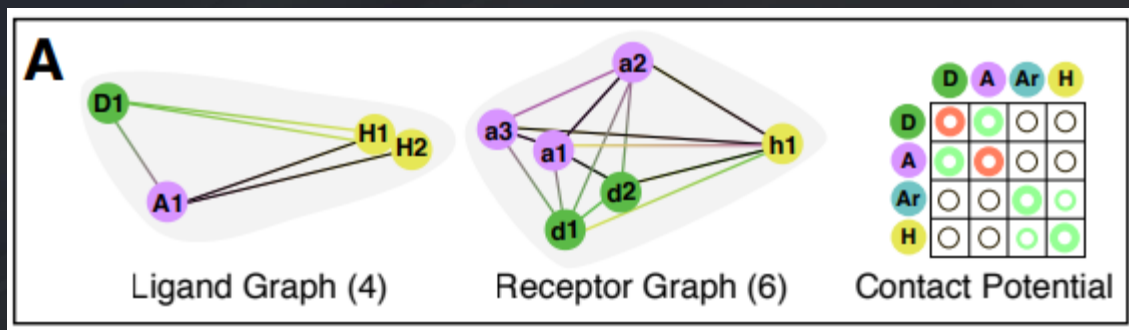
→ Identification of pharmacophore points.

→ Measure of their pairwise distance from 3D structure.

→ Labelled distance graph (B+C)

- Vertices = pharmacophore points
- edge weights = pairwise distance

Binding Interaction Graph: starting point

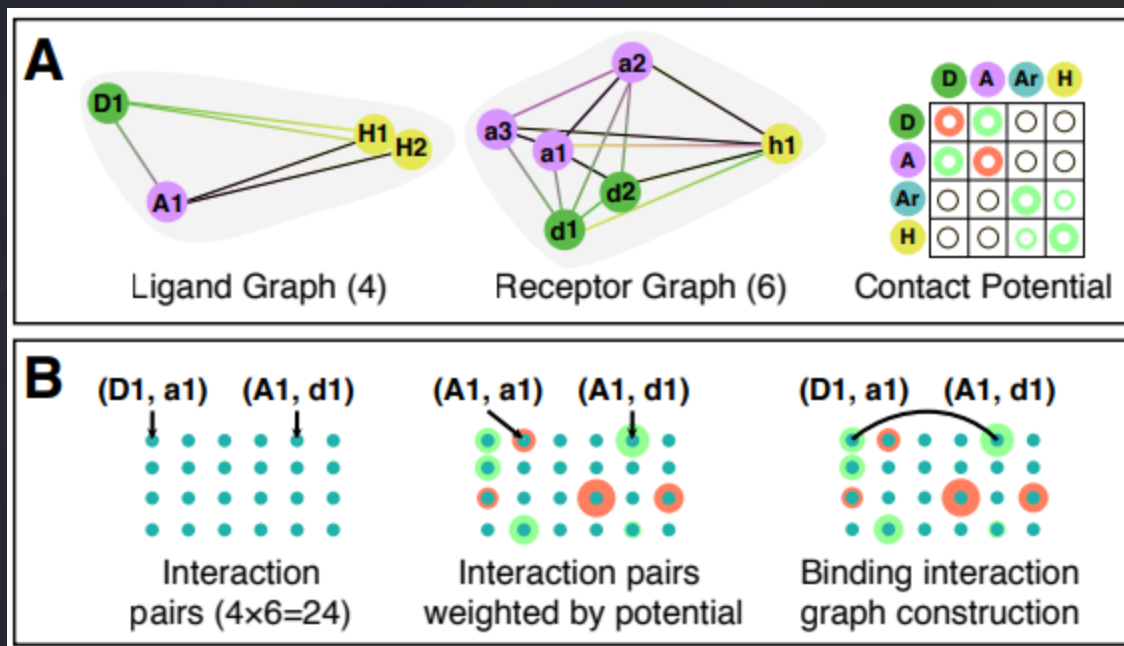


Labelled distance graphs for ligand and receptor:

- Selection of pharmacophore points
- Evaluation of contact potentials and distances

Combine them into a single Binding Interaction Graph

Binding Interaction Graph: vertices



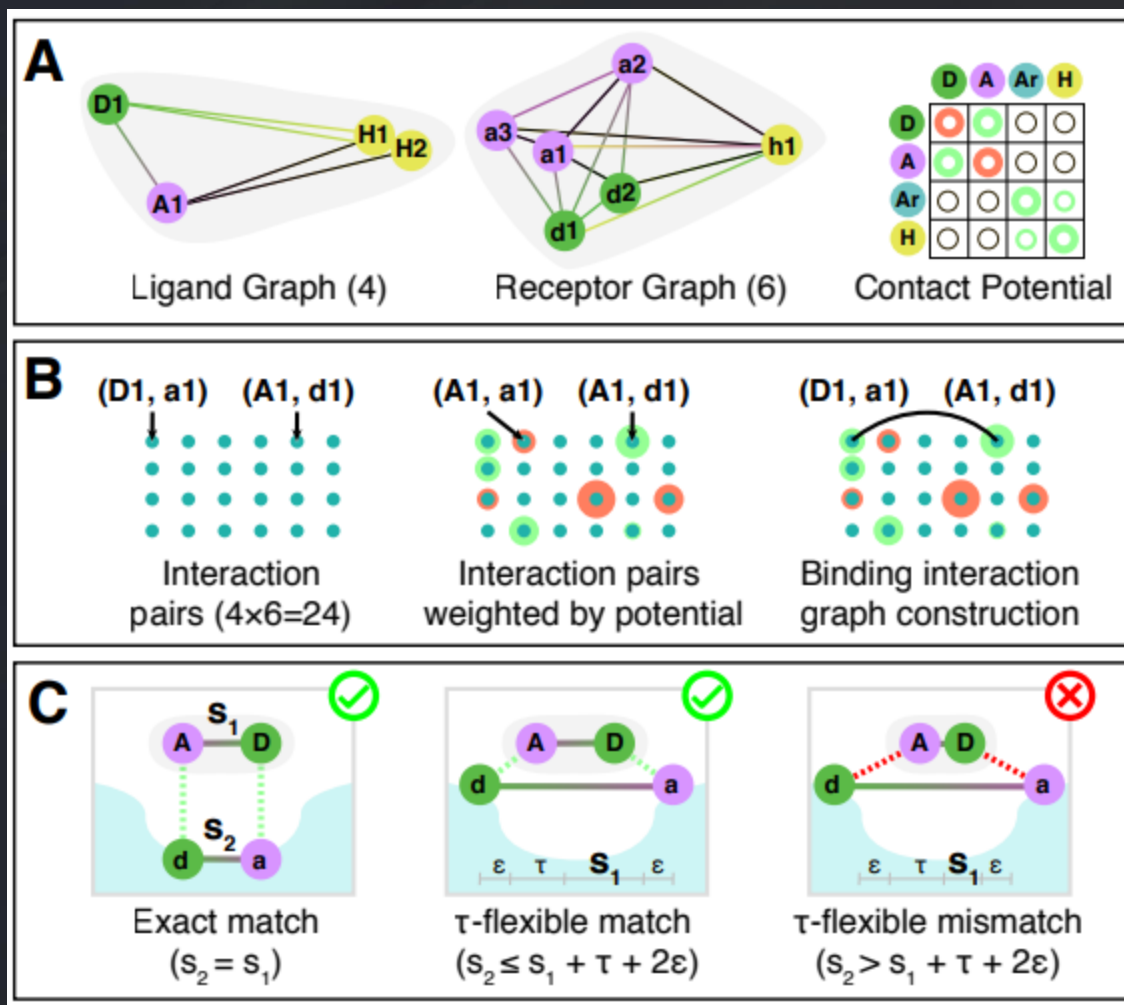
Labelled distance graphs for ligand and receptor:

- Selection of pharmacophore points
- Evaluation of contact potentials and distances

Combine them into a single Binding Interaction Graph

- A **vertex** for each possible contact between ligand (n) and the receptor (m) pharmacophore points. In principle $n \times m$ vertices.
- Each vertex is **weighted** by the contact potential

Binding Interaction Graph: edges



Labelled distance graphs for ligand and receptor:

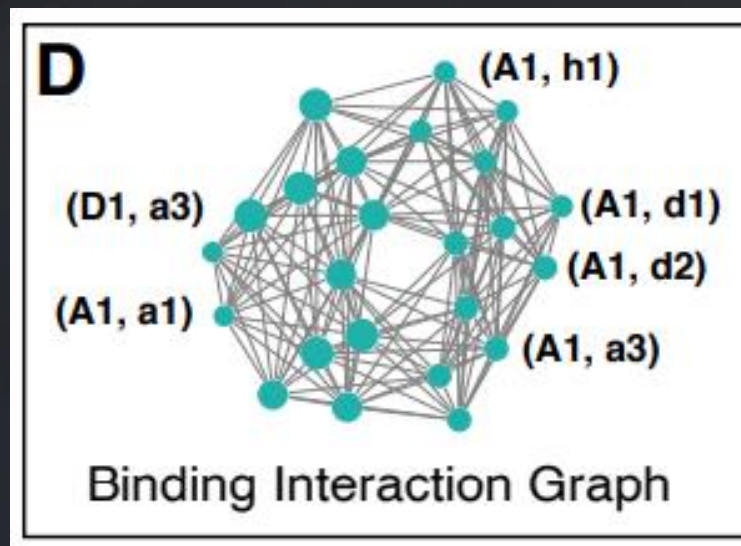
- Selection of pharmacophore points
- Evaluation of contact potentials and distances

Combine them into a single Binding Interaction Graph

- A **vertex** for each possible contact between ligand (n) and the receptor (m) pharmacophore points. In principle $n \times m$ vertices.
- Each vertex is **weighted** by the contact potential
- An **edge** connects two vertices if the two contacts they represent are compatible (i.e. similar distance)

τ = flexibility constant
 ϵ = interaction distance

Binding Interaction Graph

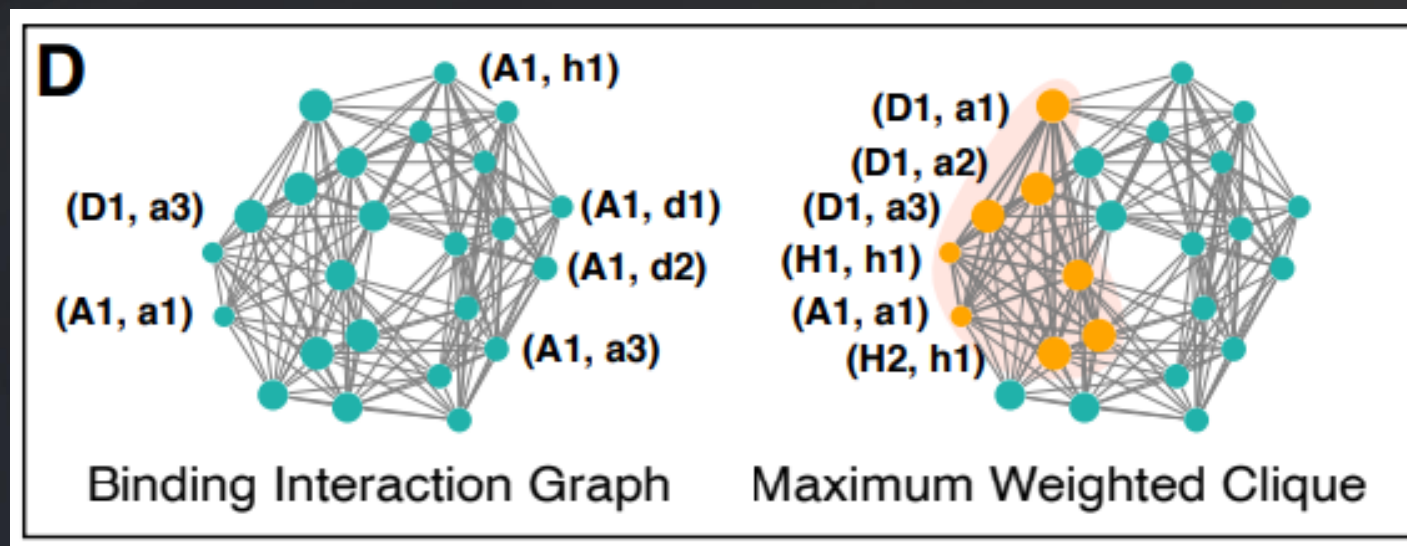


Nodes: each possible contact between ligand and the receptor

Edges connect the nodes that are τ -compatible

Weights on the nodes: contact potentials

Maximum Weighted Clique Problem (MWCP)



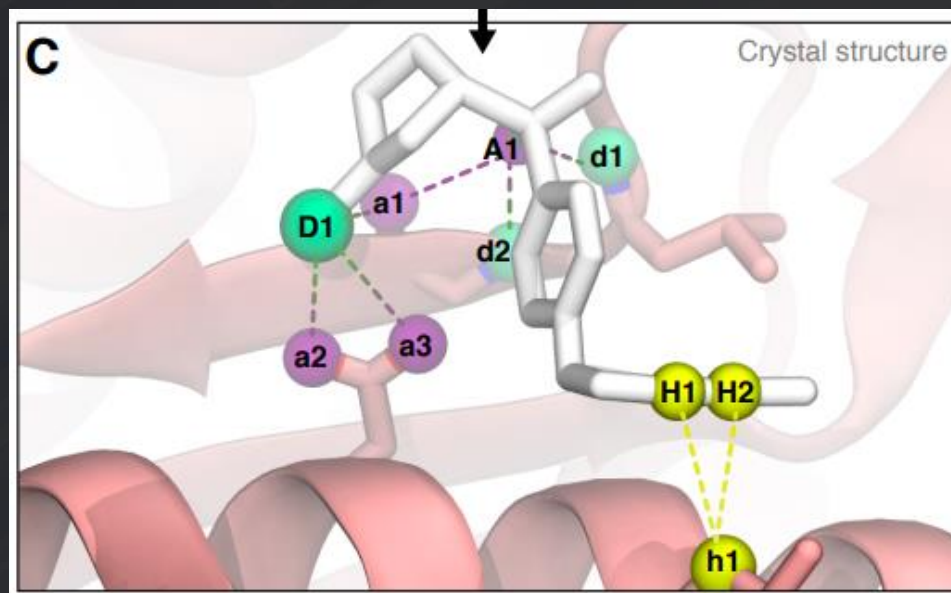
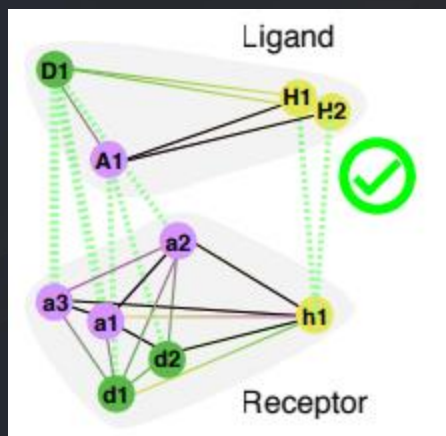
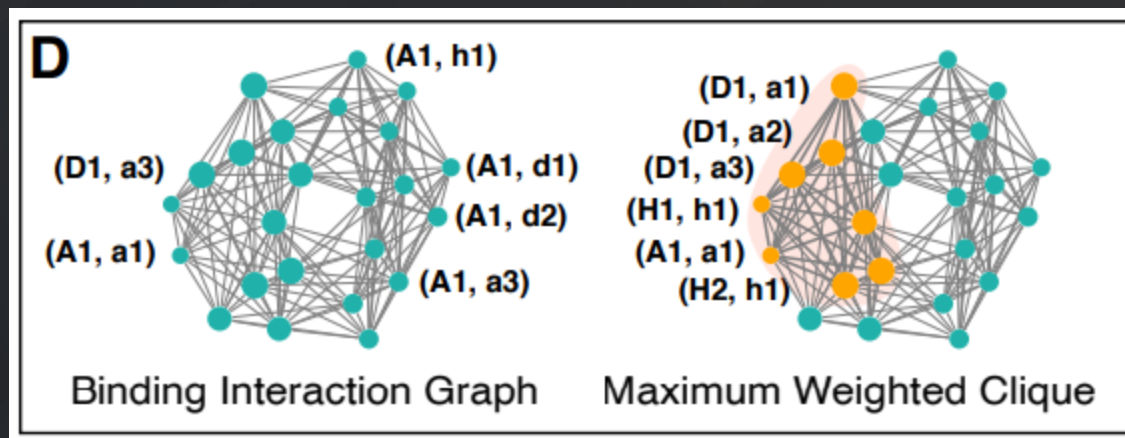
Clique: subgraph where all pairs of vertices are connected by an edge.

Maximum Clique: a clique with the largest possible number of vertices.

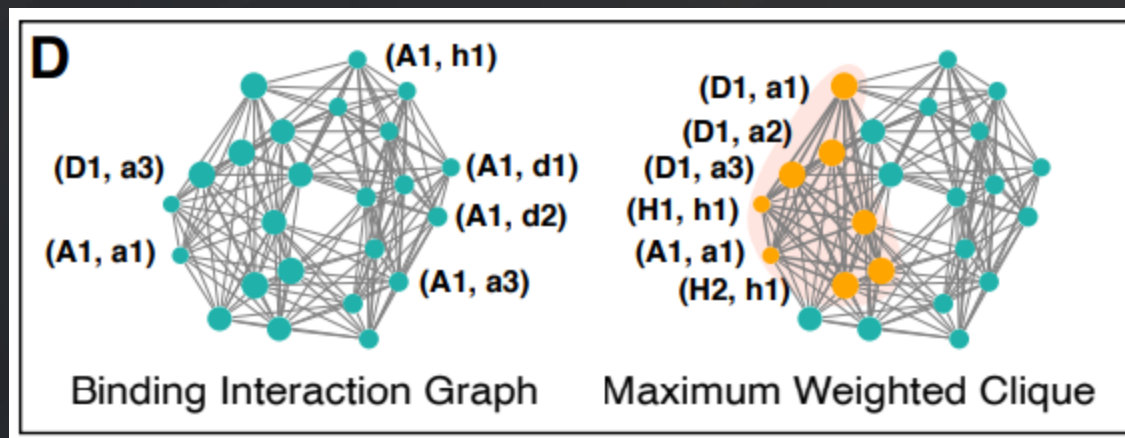
Maximum Weighted Clique (MWC): maximum clique also maximizing a cost function given by the weights.

The MWCP is a NP-hard problem

Maximum Weighted Clique Problem (MWCP)



From MWCP to MWIS

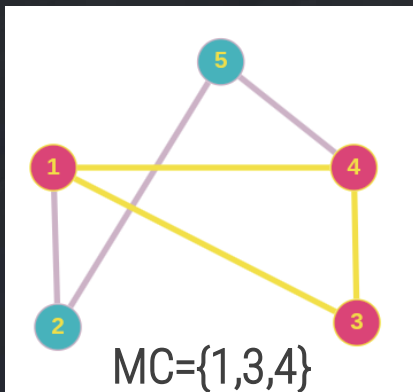


How to solve with neutral atom devices?

Find the **Maximum Weighted Independent Set (MWIS)**
on the
complement of the binding interaction graph

From MWCP to MWIS

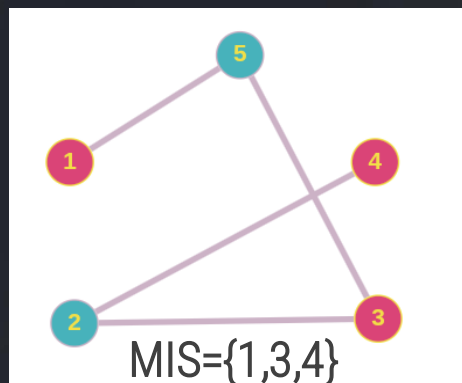
Graph



Binding Interaction Graph:

- Links indicate the *geometric compatibility* between the binding poses (nodes).

Complement

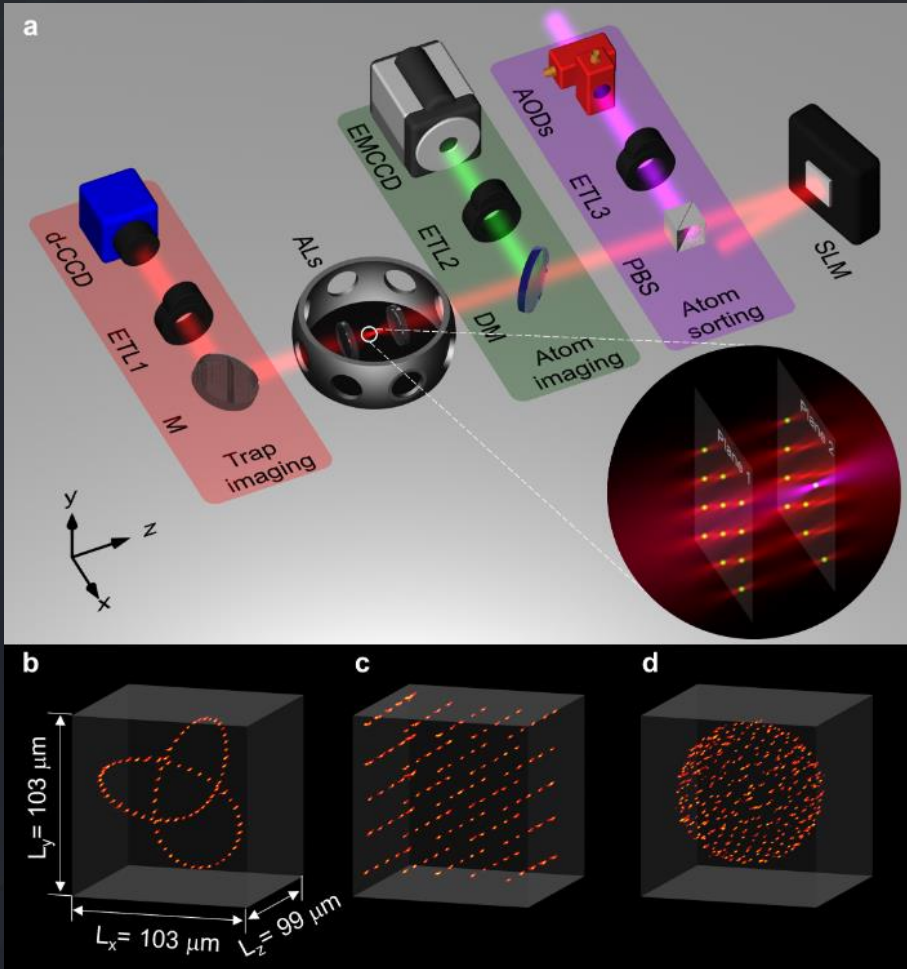


Complement of Binding Interaction Graph:

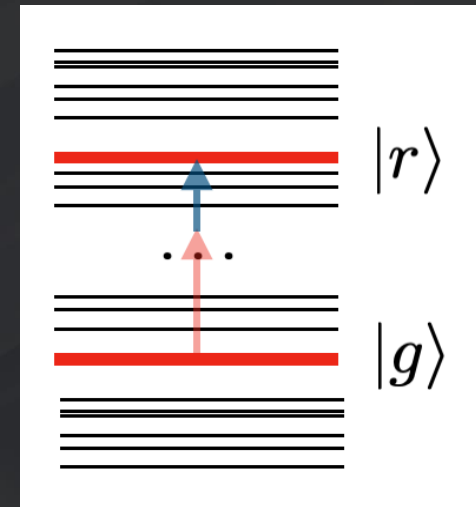
- Missing Links indicate the *geometric compatibility* between the binding poses (nodes).

Maximum independent set: largest set of nodes that are not connected to each other.

Neutral atom quantum computing



Qubit encoded in electronic states of ^{87}Rb atoms
Ground $|g\rangle$ and Rydberg $|r\rangle$



Neutral atom quantum computing

$$H = \sum_i \frac{\hbar}{2} \left(\Omega(t) \sigma_i^x - \delta(t) \sigma_i^z \right) + \sum_{i < j} U_{ij} \hat{n}_i \hat{n}_j$$

Rabi Frequency

Detuning

$$U_{ij} = \frac{C_6}{r_{ij}^6}$$

$$\hat{n}_j = (\mathbb{I} + \sigma_j^z)/2$$

Van der Waals coupling

Neutral atom quantum computing

$$H = \sum_i \frac{\hbar}{2} \left(\Omega(t) \sigma_i^x - \delta(t) \sigma_i^z \right) + \sum_{i < j} U_{ij} \hat{n}_i \hat{n}_j$$

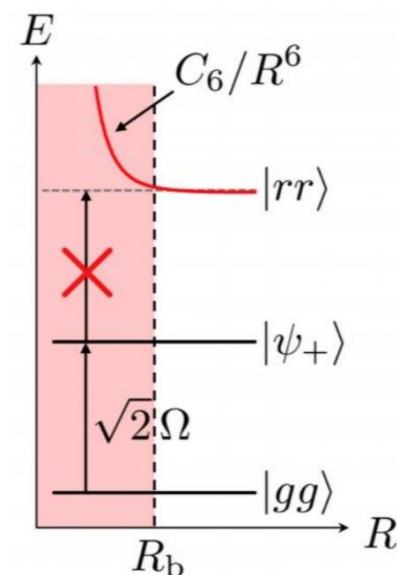
Rabi Frequency

Detuning

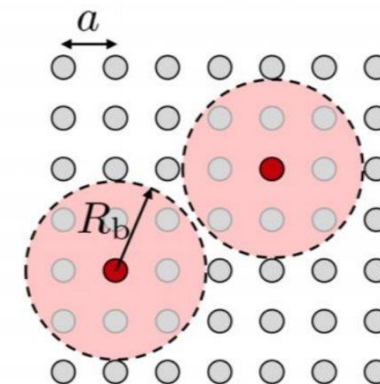
$$U_{ij} = \frac{C_6}{r_{ij}^6}$$

$$\hat{n}_j = (\mathbb{I} + \sigma_j^z)/2$$

Van der Waals coupling



Rydberg blockade



Neutral atom quantum computing

$$H = \sum_i \frac{\hbar}{2} \left(\Omega(t) \sigma_i^x - \delta(t) \sigma_i^z \right) + \sum_{i < j} U_{ij} \hat{n}_i \hat{n}_j$$

$$\hat{n}_j = (\mathbb{I} + \sigma_j^z)/2$$

Rabi Frequency

local Detuning

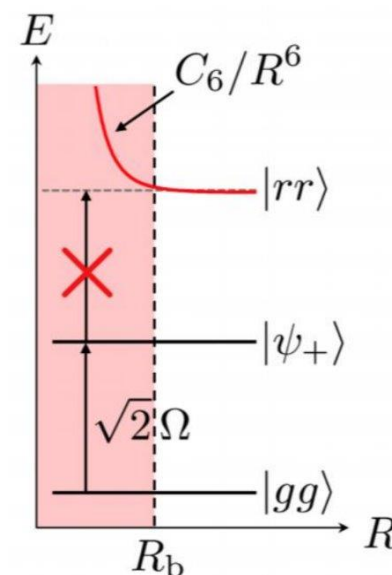
Needed for **encoding weights** in MWIS

$$U_{ij} = \frac{C_6}{r_{ij}^6}$$

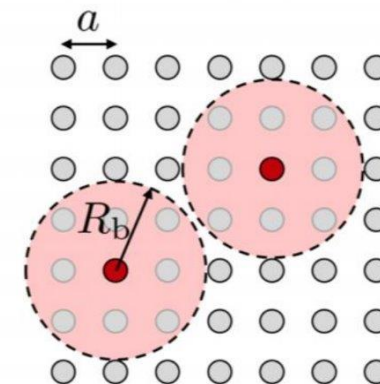
Van der Waals coupling



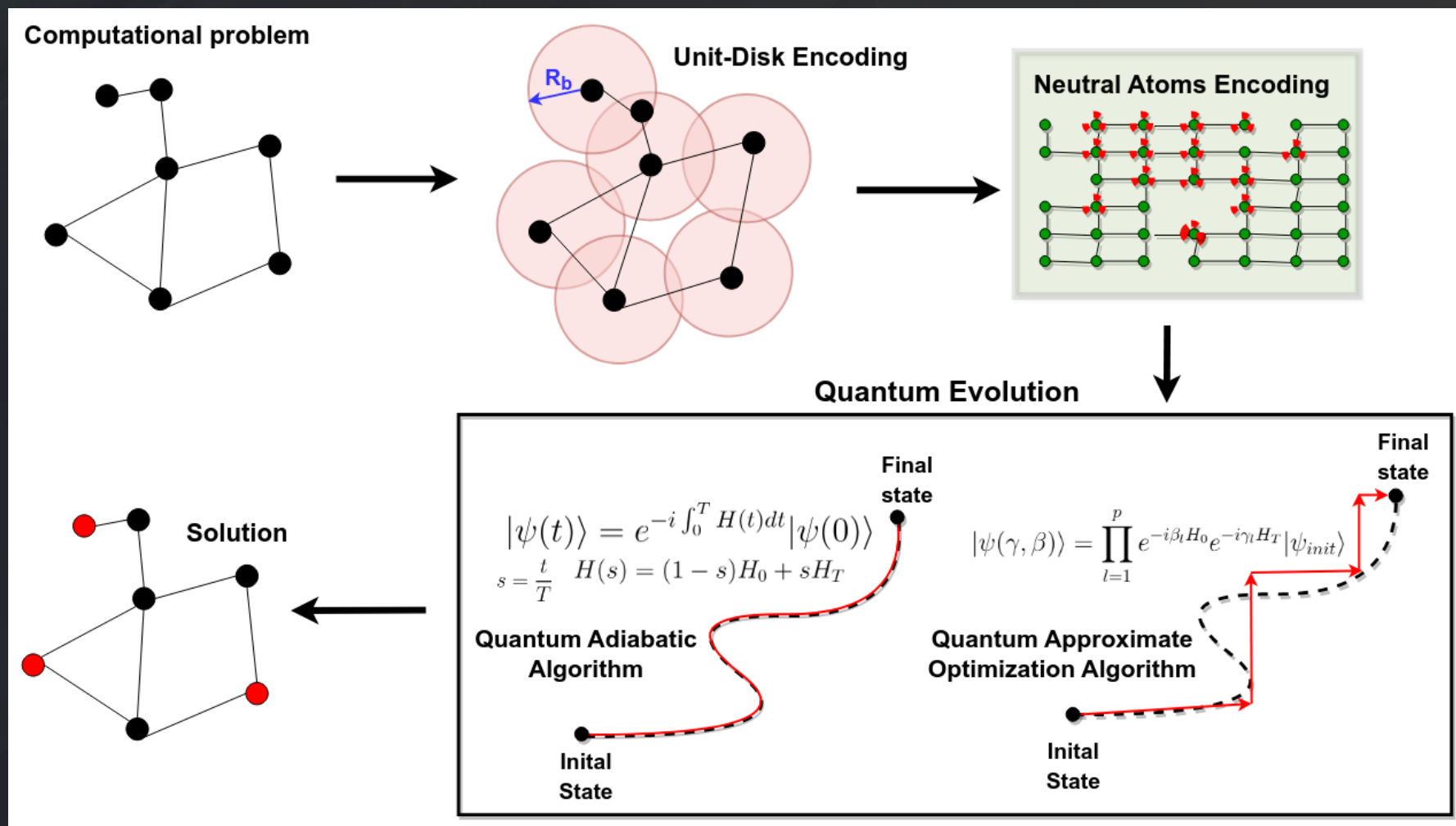
“Local addressability”
upgrade next year



Rydberg blockade

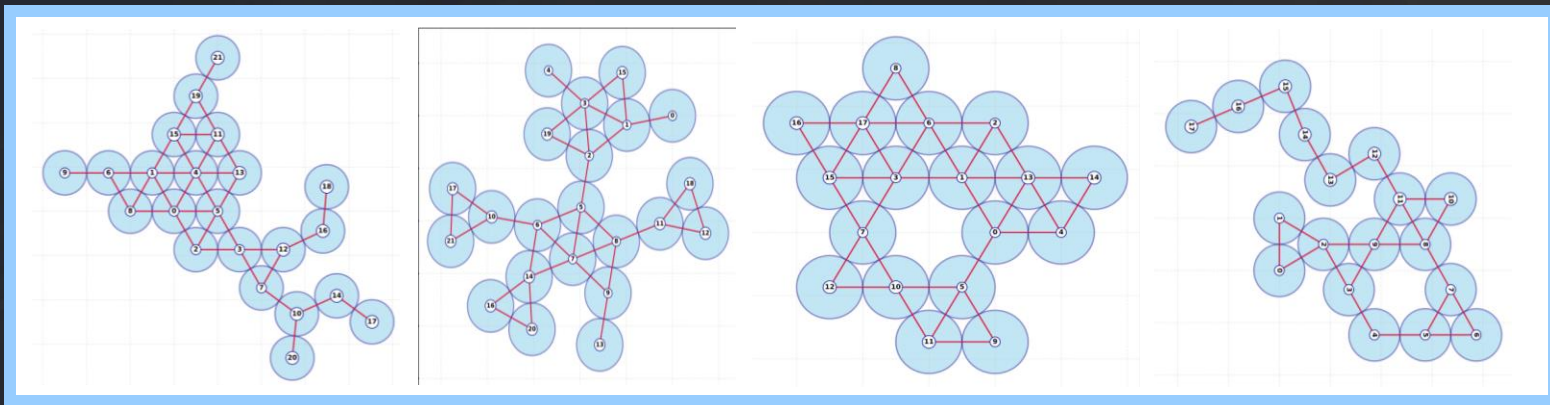


Neutral atom quantum computing



Benchmarking QAA for MWIS

Synthetic graph dataset



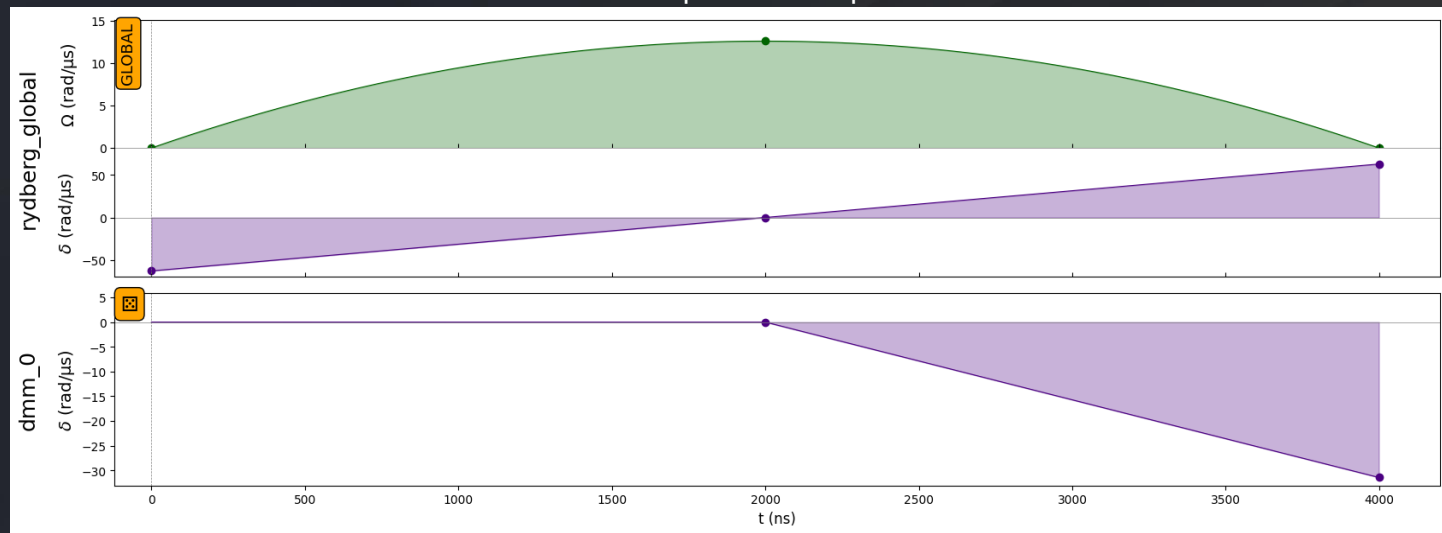
Elisa Ercolessi



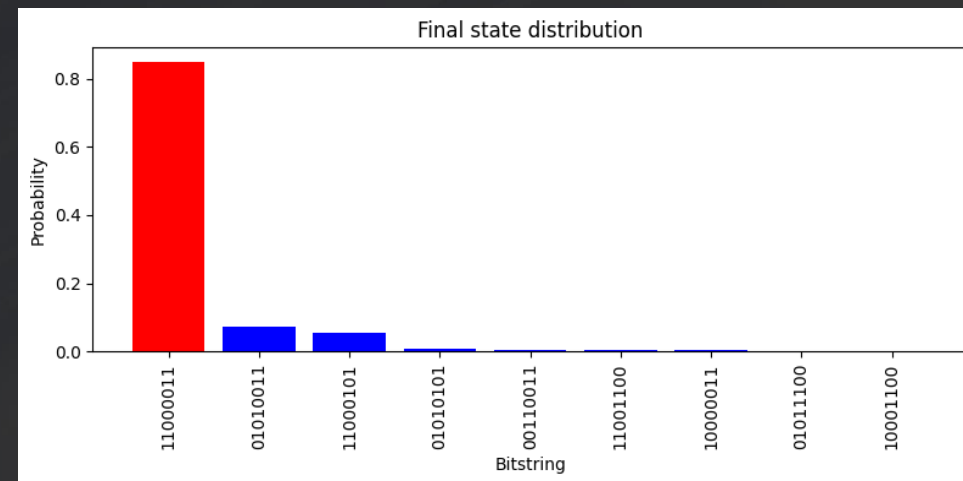
Matteo Grotti

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Adiabatic pulse sequence



Results



Benchmarking QAA for MWIS

- Finding the **optimal parameters**:
 - Inter-atomic distance
 - Pulse sequence sweep time
- Determine reasonable **weight ratios**
- Assess the impact of **noise**



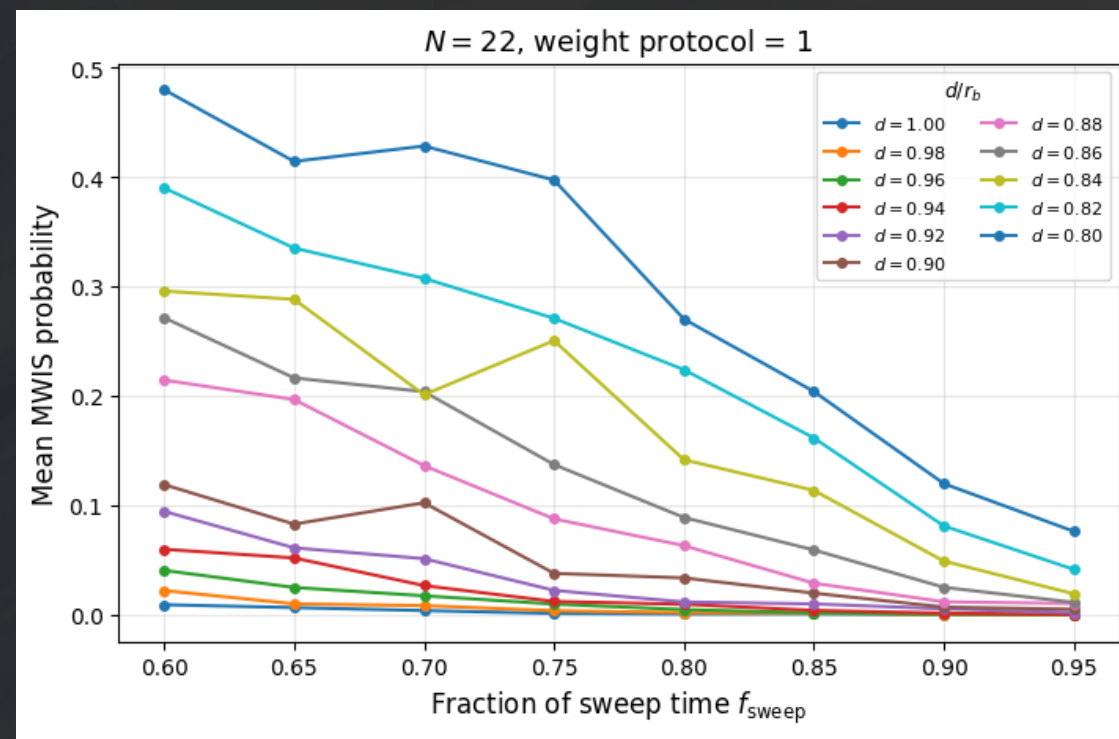
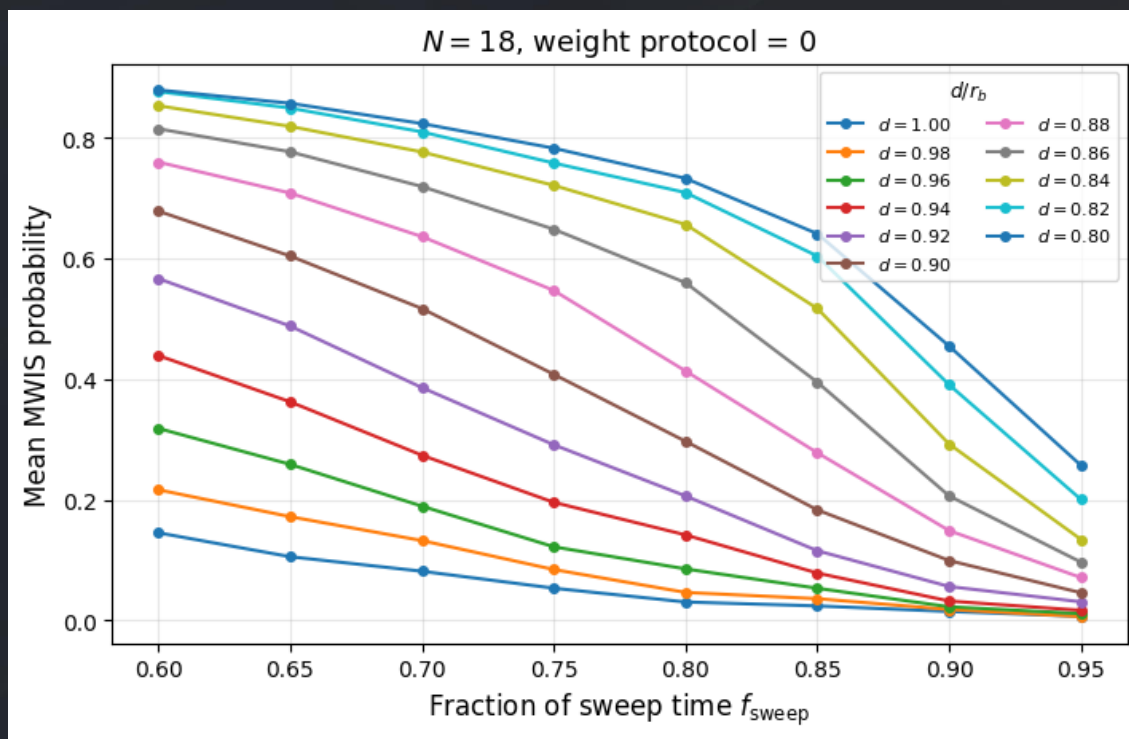
Elisa Ercolessi



Matteo Grotti

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Preliminary results





Thank you
Questions?